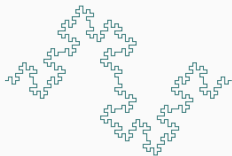


Course on Fair Allocation of Indivisible Goods

N. Maudet

2025-2026



Master ANDROIDE — Sorbonne Université

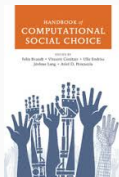
Presentation of the tutorial

The course is divided in two parts:

- fairness notions, properties
- computation, protocols for fair division

+ (time permitting) some recent topics in fair division

The content is partly based on



Download http://www.cambridge.org/download_file/932961

Presentation of the tutorial

- Jupyter Notebook accompanying the lecture available at:

`https://github.com/nmaudet/fairdiv-indivisible-items`

- Speakup for questions and polls:

`https://web.speakup.info/room/join/85567`

Other useful resources

- Other general surveys:

Chevaleyre et al. *Issues in multiagent resource allocation*. Informatica, 2006.

Lang and Rothe. *Fair Division of Indivisible Goods*. In Economics and Computation, 2016.

Moulin. *Fair Division in the Internet Age*. Annual Review of Economics, 2019.

Aziz. *Developments in Multi-Agent Fair Allocation*. AAAI-2020.

Amanatidis, Aziz, Birmpas, Filos-Ratsikas, Li, Moulin, Voudouris, Wu. *Fair division of indivisible goods: Recent progress and open questions*. Artificial Intelligence, Vol. 322, 2023.

Other useful resources

- Other general surveys:

Chevaileyre *et al.* *Issues in multiagent resource allocation*. Informatica, 2006.

Lang and Rothe. *Fair Division of Indivisible Goods*. In Economics and Computation, 2016.

Moulin. *Fair Division in the Internet Age*. Annual Review of Economics, 2019.

Aziz. *Developments in Multi-Agent Fair Allocation*. AAAI-2020.

Amanatidis, Aziz, Birmpas, Filos-Ratsikas, Li, Moulin, Voudouris, Wu. *Fair division of indivisible goods: Recent progress and open questions*. Artificial Intelligence, Vol. 322, 2023.

- Web applications

<http://www.spliddit.org/>

<https://fair-alloc.streamlit.app/>

Goldman and Procaccia. *Spliddit: Unleashing Fair Division Algorithms*. SIGecom Exchanges, 2014.

Some applications

- course allocation in Universities
<https://mba-inside.wharton.upenn.edu/course-match>
- chore allocation in households
<https://housechore-division.com/>
- food donations
- resource allocations in cloud computing
- electricity distribution

Basic Notions

Agents and Resources

- a set of **agents** $\mathcal{N} = \{1, \dots, n\}$
- a set of **resources** $\mathcal{O} = \{1, \dots, m\}$
- an **allocation** is a function $\pi : \mathcal{N} \rightarrow 2^{\mathcal{O}}$ mapping each agent to the bundle she receive.
- we denote by $\pi(i)$: bundle/share of agent i , and the set of all allocations Π

Agents and Resources

- a set of **agents** $\mathcal{N} = \{1, \dots, n\}$
- a set of **resources** $\mathcal{O} = \{1, \dots, m\}$
- an **allocation** is a function $\pi : \mathcal{N} \rightarrow 2^{\mathcal{O}}$ mapping each agent to the bundle she receive.
- we denote by $\pi(i)$: bundle/share of agent i , and the set of all allocations Π

We assume in this tutorial that **resources**:

- cannot be divided (otherwise: cake-cutting literature),
- cannot be copied, ie. $\pi(i) \cap \pi(j) = \emptyset$, for all $i, j \in \mathcal{N}$

Unless stated otherwise, we also rule out the possibility of special divisible resource (**money**).

Preferences

In this tutorial we shall mainly consider preferences that are:

- **additive**: the utility enjoyed for a bundle is simply the sum of the goods taken separately

$$u_i(B) = \sum_{r \in B} u_i(\{r\})$$

- **normalized**: additive, and the value of the full bundle is the same for all agents

$$u_i(\mathcal{O}) = K, \forall i \in \mathcal{N}$$

Preferences

In this tutorial we shall mainly consider preferences that are:

- **additive**: the utility enjoyed for a bundle is simply the sum of the goods taken separately

$$u_i(B) = \sum_{r \in B} u_i(\{r\})$$

- **normalized**: additive, and the value of the full bundle is the same for all agents

$$u_i(\mathcal{O}) = K, \forall i \in \mathcal{N}$$

- **Borda**: agents (strictly) rank resources and assign utility m to their preferred resource, $m - 1$ to their second preferred, etc.



An example

				
agent 1	5	47	45	3
agent 2	45	5	48	2
agent 3	23	25	32	20

An example

				
agent 1	5	47	45	3
agent 2	45	5	48	2
agent 3	23	25	32	20

 What is the fairest allocation according to you?

A: (1: , 2: , 3:  )

B: (1: , 2: , 3:  )

C: (1: , 2: , 3:  )

D: (1: , 2:  , 3: )

Herreiner and Puppe. *Distributing Indivisible Goods Fairly: Evidence from a Questionnaire Study*. Analyse & Kritik, 2007.

Efficiency

Although we are mainly concerned with fairness notions, efficiency requirements are still important, otherwise we may promote rather radical solutions:

Throw these candies if you can't decide how to share it!

- **completeness**: all resources must be allocated:

$$\cup_{i \in \mathcal{N}} \pi(i) = \mathcal{O}$$

- **completeness**: all resources must be allocated:

$$\cup_{i \in \mathcal{N}} \pi(i) = \mathcal{O}$$

- **Pareto-optimality (PO)**: no other allocation is as good for everyone (and strictly better for at least one agent)

- **completeness**: all resources must be allocated:

$$\cup_{i \in \mathcal{N}} \pi(i) = \mathcal{O}$$

- **Pareto-optimality (PO)**: no other allocation is as good for everyone (and strictly better for at least one agent)
- **utilitarian social welfare (USW)**: maximize the sum of agents' utilities

$$\sum_{i \in \mathcal{N}} u_i(\pi(i))$$

[Observe that completeness only refers to allocation,
Pareto-optimality is ordinal, utilitarian social welfare is cardinal]

Fairness Measures and Criteria

Different ways to assess how fair is an allocation have been proposed in the literature.

- some of them are measures, which can be optimized,
- others are boolean criteria, which are satisfied or not.

The **egalitarian social welfare** tries to maximize the utility of the agent who is the worst-off.

$$\min_{i \in \mathcal{N}} u_i(\pi(i))$$

An allocation maximizing this value **egalitarian-optimal**.

The **leximin ordering** generalizes egalitarian social welfare: agents are ranked from poorest to richest, from 1 to n . An allocation π **leximin-dominates** π' if $\pi(i) = \pi'(i)$ for all $i < j$, and $\pi(j) > \pi'(j)$.

An allocation not leximin-dominated is **leximin-optimal**.

The **proportional fair share** of an agent is the n^{th} of the utility she assigns to the full bundle

$$pfs(i) = \frac{1}{n} u_i(\mathcal{O})$$

An allocation π is **proportional** if $u_i(\pi(i)) \geq pfs(i)$, for all $i \in \mathcal{N}$

Steinhaus. *The Problem of Fair Division*. Econometrica-1948.

The **maxmin fair share** of an agent is the best share she can guarantee herself in a game “I cut, I choose last”.

$$mfs(i) = \max_{\pi \in \Pi} \min_{j \in \mathcal{N}} u_i(\pi(j))$$

An allocation π satisfies maxmin fair share if $u_i(\pi(i)) \geq mfs(i)$, for all $i \in \mathcal{N}$

Budish. *The combinatorial assignment problem*. Journal of Political Economy. 2011.

An allocation is **envy-free** when no agent prefers the bundle of another agent over her own bundle.

$$u_i(\pi(i)) \geq u_i(\pi(j)), \text{ for all } i, j \in \mathcal{N}$$

Foley. *Resource allocation and the public sector*. Yale Econ Essays. 1967.

Fairness as equilibrium search:

- each object has a public price
- each agent has the same budget

A pair (π, p) is CEEI if, for each agent, her share is maximal among the ones which could be “buy” given the public prices of objects.

An allocation π satisfies CEEI if there exists a price vector p such (π, p) is CEEI.

Example

Two agents, five resources, additive utilities.

					
agent 1	6	6	6	0	0
agent 2	5	5	3	3	2

Example

Two agents, five resources, additive utilities.

						<i>pfs</i>	<i>mfs</i>
agent 1	6	6	6	0	0	$18/2 = 9$	6
agent 2	5	5	3	3	2	$18/2 = 9$	8

Example

Two agents, five resources, additive utilities.

						<i>pfs</i>	<i>mfs</i>
agent 1	6	6	6	0	0	$18/2 = 9$	6
agent 2	5	5	3	3	2	$18/2 = 9$	8

- find an allocation satisfying MFS but not PROP

Example

Two agents, five resources, additive utilities.

						<i>pfs</i>	<i>mfs</i>
agent 1	6	6	6	0	0	$18/2 = 9$	6
agent 2	5	5	3	3	2	$18/2 = 9$	8

- find an allocation satisfying MFS but not PROP
- does there exist an EF allocation?

Example

Two agents, five resources, additive utilities.

						<i>pfs</i>	<i>mfs</i>
agent 1	6	6	6	0	0	$18/2 = 9$	6
agent 2	5	5	3	3	2	$18/2 = 9$	8

- find an allocation satisfying MFS but not PROP
- does there exist an EF allocation?
- what is the egalitarian social welfare for this problem?

Example

Two agents, five resources, additive utilities.

						<i>pfs</i>	<i>mfs</i>
agent 1	6	6	6	0	0	$18/2 = 9$	6
agent 2	5	5	3	3	2	$18/2 = 9$	8

- find an allocation satisfying MFS but not PROP
- does there exist an EF allocation?
- what is the egalitarian social welfare for this problem?
- is there an allocation such that both agents would be envious?

Example

Two agents, five resources, additive utilities.

						<i>pfs</i>	<i>mfs</i>
agent 1	6	6	6	0	0	$18/2 = 9$	6
agent 2	5	5	3	3	2	$18/2 = 9$	8

- find an allocation satisfying MFS but not PROP
- does there exist an EF allocation?
- what is the egalitarian social welfare for this problem?
- is there an allocation such that both agents would be envious?
- check there exists an allocation CEEI with price vector:
 $\langle 0.4, 0.5, 0.5, 0.3, 0.3 \rangle$

The envy of i towards j can be eliminated by removing **some** resource from j 's bundle (“envy up to the most envied good”).

$$\forall i, j \in \mathcal{N} \exists r \in \pi(j) : u_i(\pi(i)) \geq u_i(\pi(j) \setminus \{r\})$$

A stronger version requiring that the removal of **any** resource eliminates envy (“envy up to the least envied good”):

$$\forall i, j \in \mathcal{N} \forall r \in \pi(j) : u_i(\pi(i)) \geq u_i(\pi(j) \setminus \{r\})$$

Relaxation: Degrees of Envy

It is possible to define various notions of degrees of envy, by combining operators at different levels:

- envy **between two agents** : we may consider

$$e_{ij} = \max(u_i(\pi(j)) - u_i(\pi(i)), 0)$$

Relaxation: Degrees of Envy

It is possible to define various notions of degrees of envy, by combining operators at different levels:

- envy **between two agents** : we may consider

$$e_{ij} = \max(u_i(\pi(j)) - u_i(\pi(i)), 0)$$

- degree of envy of a **single agent** i:

$$\max_{j \in \mathcal{N}} e_{ij}, \text{ or } \sum_{j \in \mathcal{N}} e_{ij}$$

Relaxation: Degrees of Envy

It is possible to define various notions of degrees of envy, by combining operators at different levels:

- envy **between two agents** : we may consider

$$e_{ij} = \max(u_i(\pi(j)) - u_i(\pi(i)), 0)$$

- degree of envy of a **single agent** i:

$$\max_{j \in \mathcal{N}} e_{ij}, \text{ or } \sum_{j \in \mathcal{N}} e_{ij}$$

- degree of envy of **the society**:

$$\max_{i \in \mathcal{N}} e_i, \text{ or } \sum_{i \in \mathcal{N}} e_i$$

Chevaleyre, Endriss, M. *Distributed Fair Allocation of Indivisible Goods*. Artificial Intelligence, 2017.

Relaxation: Degrees of Envy

				
agent 1	5	47	45	3
agent 2	45	5	48	2
agent 3	23	25	32	20

Consider this (bad) allocation... A : (1 : , 2 : , 3 : )

Envy Matrix

	1	2	3	max
agent 1	0	45	40	45
agent 2	38	0	41	41
agent 3	0	13	0	13
max				45

Relaxation: Degrees of Envy

				
agent 1	5	47	45	3
agent 2	45	5	48	2
agent 3	23	25	32	20

Consider this (bad) allocation... A : (1 : , 2 : , 3 : )

Envy Matrix

	1	2	3	max
agent 1	0	45	40	45
agent 2	38	0	41	41
agent 3	0	13	0	13
max				45

... Or perhaps we could balance the envy vector of agents?

Relaxation: Proportionality up to some/any good PROP1/X

An allocation π is **PROP1** if $u_i(\pi(i) \cup \{r\}) \geq pfs(i)$, for some good r not in $\pi(i)$ (“up to the most valued good”), for all $i \in \mathcal{N}$

An allocation π is **PROPX** if $u_i(\pi(i) \cup \{r\}) \geq pfs(i)$, for any good r not in $\pi(i)$ (“up to the least valued good”), for all $i \in \mathcal{N}$

Conitzer, Freeman, Shah. *Fair Public Decision Making*. EC-2017.

Aziz, Moulin, and Sandomirskiy. *A polynomial-time algorithm for computing a Pareto optimal and almost proportional allocation*. Operations Research Letters, 2020.

See also the recently proposed intermediary notion of **PROP-m**, proportionality up to the maximin item (guaranteed to exist).

Baklanov et al. *Achieving Proportionality up to the Maximin Item with Indivisible Goods*. AAAI-21.

Baklanov et al. *PROPM Allocations of Indivisible Goods to Multiple Agents*. IJCAI-21.

Relaxation: Envy under Restricted Knowledge

Envy relies on the fact that agents have full knowledge of the allocation. Other assumptions are possible (and make sense in many settings):

- agents can only see some other agents (underlying social network): **Local Envy-Freeness**
- agents have only knowledge about the available items and the number of agents: **Epistemic Envy-Freeness**

Note: possible to mix some of these relaxations.

In particular, Epistemic Envy-Freeness and Envy up to any good (EEFX) is guaranteed to exist.

Relation between measures

With additive valuations, are EF allocations always PROP?

✍ A : yes B : no

Relation between measures

With additive valuations, are EF allocations always PROP?

 A : yes B : no

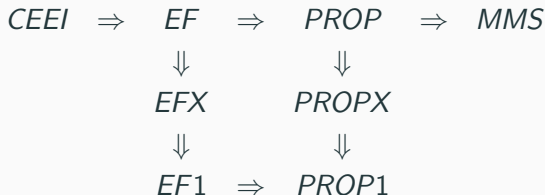
With additive valuations, EF allocations always PROP

Suppose for contradiction that this is not the case.

- We have an allocation EF and not PROP, hence it must be that there is an agent i s.t. $u_i(\pi(i)) < \frac{1}{n}u_i(\mathcal{O})$
- thus $u_i(\mathcal{O} \setminus \pi(i)) > \frac{(n-1)}{n} \cdot u_i(\mathcal{O})$
- but then at least one of the $(n-1)$ agents must hold a bundle that agent i values more than $\frac{1}{n}u_i(\mathcal{O})$. Contradiction.

The landscape of fairness measures

For additive preferences, the following scale of criteria holds:



Bouveret, Lemaitre. *Characterizing Conflicts in Fair Division of Indivisible Goods Using a Scale of Criteria*. JAAMAS-2014.

Conitzer, Freeman, Shah. *Fair Public Decision Making*. EC-2017.

Aziz, Moulin, and Sandomirskiy. *A polynomial-time algorithm for computing a Pareto optimal and almost proportional allocation*. Operations Research Letters, 2020.

Existence: What is known... and not known

It is obvious to see that PROP (hence EF and CEEI) may not exist. (Think a of single good to be allocated among two agents)

MMS may not exist / PROPX may not exist

Procaccia and Wang. *Fair enough: Guaranteeing approximate maximin shares*. EC-2014.

Moulin. *Fair Division in the Internet Age*. Annual Review of Economics, 2019.

Open question: does an EFX always exist?

Plaut, Roughgarden. *Almost envy-freeness with general valuations*. SODA-2018.

Chaudhury, Garg, Mehlhorn. *EFX Exists for Three Agents*. EC-2020.

Procaccia. *An answer to fair division's most enigmatic question: technical perspective*. In: Comm. ACM 2020.

EF1 (hence PROP1) always exist

Fairness + Efficiency

On the relation between fairness and efficiency

Are EF allocation necessarily Pareto-optimal?

 A : yes B : no

On the relation between fairness and efficiency

Are EF allocation necessarily Pareto-optimal?

 A : yes B : no

EF allocation are not necessarily Pareto-optimal

Here is a counter-example:

				
agent 1	10	0	1	2
agent 2	0	10	2	1

On the relation between fairness and efficiency

Are EF allocation necessarily Pareto-optimal?

 A : yes B : no

EF allocation are not necessarily Pareto-optimal

Here is a counter-example:

				
agent 1	10	0	1	2
agent 2	0	10	2	1

Price of Fairness

As there is usually no simple relation between efficiency and fairness measures, a natural question is thus to study a notion of “cost” of fairness:

$$\frac{\text{utilitarian opt}}{\text{utilitarian sw of best allocation satisfying fairness}}$$

This is known as the **price of fairness**.

Caragiannis et al. *The efficiency of fair division*. TOCS-2012.

Price of Fairness

Eg., what is the price to pay if we insist on the allocation to be ESW-optimal?

Consider the following example (with additive utilities).

					
agent 1	ϵ	$1 - \epsilon$	0	0	0
agent 2	0	ϵ	$1 - \epsilon$	0	0
agent 3	0	0	ϵ	$1 - \epsilon$	0
agent 4	0	0	0	ϵ	$1 - \epsilon$
agent 5	0	0	0	0	1

Price of Fairness

Eg., what is the price to pay if we insist on the allocation to be ESW-optimal?

Consider the following example (with additive utilities).

					
agent 1	ϵ	$1 - \epsilon$	0	0	0
agent 2	0	ϵ	$1 - \epsilon$	0	0
agent 3	0	0	ϵ	$1 - \epsilon$	0
agent 4	0	0	0	ϵ	$1 - \epsilon$
agent 5	0	0	0	0	1

$$\Rightarrow \text{PoF} = \frac{(n-1)(1-\epsilon)}{1+(n-1)\cdot\epsilon} = n$$

Back to our initial example

				
agent 1	5	47	45	3
agent 2	45	5	48	2
agent 3	23	25	32	20

 What is the fairest allocation according to you? Results!

Alloc.		You	HP	Prop.
A :	(1 :  , 2 :  , 3 :  )	44	26.2	ESW, IA
B :	(1 :  , 2 :  , 3 :  )	22	12.7	ESW, PO
C :	(1 :  , 2 :  , 3 :  )	22	26.2	EF, PO
D :	(1 :  , 2 :   , 3 : )	12	9	USW, PO

Mechanisms

We first address the question of the computation, by a central authority, of the notion mentioned earlier.

In particular, we will have a look at :

- computing ESW-optimal (ie. maxmin) allocation
- computing envy-free allocations

Computing maxmin allocations

In general, the problem is computationally **difficult** (NP-hard): it is likely that no polynomial algorithm can solve this problem.

What is perhaps more surprising is that the problem remains difficult even if agents have additive utilities.

This problem has been studied as the **Santa Claus problem**.

[By comparison, observe that computing an utilitarian-optimal allocation is easy in that case.]

Bansal and Sviridenko. *The Santa Claus problem*. STOC-2006.

The assignment MIP

Let us make use of binary variables x_{ij} to express that agent i holds resource r_j ($=1$), or not ($=0$)

$$\begin{array}{ll}\text{maximize} & y \\ \text{subject to} & x_{i,j} \in \{0, 1\}, \quad j = 1, \dots, m \\ & i = 1, \dots, n \\ & \sum_{i \in N} x_{ij} = 1, \quad j = 1, \dots, m \\ & \sum_{r_j \in O} u_i(r_j) \times x_{ij} \geq y, \quad i = 1, \dots, n\end{array}$$

Computing envy-free allocation is...

Computing envy-free allocation is...
simple if we don't impose any efficiency requirement.

Even with additive utilities, deciding whether there exists:

- a **complete EF** allocation is NP-complete,
- a **PO + EF** allocation is even higher in the hierarchy.

Lipton et al. *On approximately fair allocations of indivisible goods*. EC-04.

de Keijzer et al. *On the complexity of efficiency and envy-freeness in fair division of indivisible goods with additive preferences*. ADT-09.

The envy-minimizing MIP

We can express

$$\begin{array}{ll}\text{minimize} & y \\ \text{subject to} & x_{ij} \in \{0, 1\}, \quad j = 1, \dots, m \\ & i = 1, \dots, n \\ & \sum_{i \in N} x_{ij} = 1, \quad j = 1, \dots, m \\ & \sum_{r_j \in O} u_i(r_j) \times x_{i'j} - \sum_{r_j \in O} u_i(r_j) \times x_{ij} \leq y, \quad i, i' = 1, \dots, n \\ & i' \neq i\end{array}$$

Many more results...

- Similar results have been obtained for many more fairness measures.

Furthermore, to circumvent the difficulty of many of these problems, it is possible to study:

- **approximation** algorithms, which would return solution with worst-case guarantees wrt. optimal solutions,
- study **specific cases**, in terms of preference structures, number of agents, number of resources, etc.

Nguyen, Roos, Rothe. *A survey of approximability and inapproximability results for social welfare optimization in multiagent resource allocation*. AMAI13.

Interlude: How likely is it to get an EF allocation?

Intuitively, the more resources we get, the easier it should be to get an EF allocation.

Can we get a more precise statement about that?

Interlude: How likely is it to get an EF allocation?

Intuitively, the more resources we get, the easier it should be to get an EF allocation.

Can we get a more precise statement about that?

(Dickerson *et al.*) studied this for **uniform** and **correlated** utilities (in fact for more general distribution satisfying some axioms), getting asymptotic results and experimental evidence.

- the number of items needed on top of n to ensure envy-free is linear in n
- a phase transition phenomena is observed

 Let's try this on the notebook

Dickerson *et al.* *The computational rise and fall of fairness*. AAAI-14.

Maximizing Nash social welfare

The **Nash social welfare** tries to maximize the product of utilities of the agents.

$$\prod_{i \in \mathcal{N}} u_i(\pi(i))$$

An allocation maximizing this value is **Nash-optimal**.

Maximizing Nash social welfare

“The Nash solution exhibits an elusive combination of fairness and efficiency properties, and can be easily computed in practice. It provides the most practicable approach to date (arguably, the ultimate solution) for the division of indivisible goods under additive valuations.”

Maximizing NSW guarantees EF1 and PO, and it provides a theoretical guarantee on an approximation of MMS

But recall it may fail to return an EF allocation even if one exists
The complexity of finding a EF1 + PO remains an open problem

Caragiannis et al.. *The Unreasonable Fairness of Maximum Nash Welfare*. EC-2016.

Barman, Krishnamurthy, Vaish. *Finding Fair and Efficient Allocations*. EC-2018.